

N I T, Kurukshetra
(Theory Examination)
Question Paper

May/June, 2026
B Tech, 4th Semester
Subject: Database Management Systems
Course No: CSPC-206/DSPC-206

Max. Marks: 100
Time Allowed: 3 hrs

- Note: (i) Attempt all questions.
(ii) The question paper is of 100 marks; however, its weight is 50. Therefore, marks obtained by a student in this theory paper will be halved in calculating final grade.

- Q1 (a) Draw a block diagram of the DBMS.
(b) What is weak entity? How will you convert it into a strong entity? Give an example.
(c) Whether every candidate key will necessarily be a super key? How do you make a candidate key from a given super key? Give an example.
(d) A database designer decides to store a student's age instead of date of birth in the database.
(e) Is it a good design decision? If not, why? **NO**
Write a mathematical relationship between domains of primary and foreign key attributes.
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- Q2 (a) Design an ER schema for a hospital management system. The requirements are as under
- A patient may visit many doctors. **m:n**
 - A doctor treats many patients. **1:n**
 - Every appointment has date, time and diagnosis. **1:n**
 - Every doctor belongs to one department. **1:n**
 - A department has many doctors. **1:n**
 - Patients may be admitted to rooms. **1:n**
- Identify entity types with their attributes, relationship types, and draw ER schema diagram

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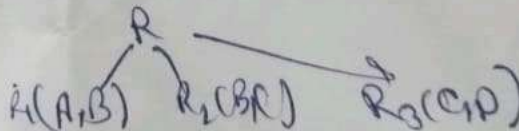
- (b) Consider the relation
STUDENT (RollNo, Name, Dept, HOD)
The following functional dependencies hold:
 $\text{RollNo} \rightarrow \text{Name, Dept}$
 $\text{Dept} \rightarrow \text{HOD}$
Find all candidate keys. **RollNo**
Identify the highest normal form. **2NF**
Explain whether transitive dependency exists? If yes how? **Yes Dept $\rightarrow$ HOD**

- Q3 (a) Given relation R(A,B,C,D) with FDs

$A \rightarrow B$
 $B \rightarrow C$
 $C \rightarrow D$

Find

1. Candidate Key **A**
2. BCNF Decomposition



- (b) Draw state diagram of a transaction.
(c) What is starvation in transactions? When does it occur? What is a simple technique to overcome it?

Q4 (a) Consider the following schedule involving two transactions:

Step	Operation
1	$R_1(X)$
2	$W_2(X)$
3	$R_2(Y)$
4	$W_1(X)$
5	$W_2(Y)$
6	$R_1(Y)$

Draw the precedence graph. Determine whether the schedule is conflict serializable? If serializable, give equivalent serial schedule.

Not

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(b) What is cascading rollback? Under which condition a schedule is cascadeless?

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Q5 (a) What is deadlock in lock-based protocols? Suggest a simple technique to overcome deadlock.

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(b) Explain timestamp ordering protocol.

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